Professional Self-Assessment

Completing my Bachelor of Science in Computer Science has given me a much better understanding of what goes into developing software and solving problems with technology. When I first started the program, I knew I was interested in computers, but I was still figuring out what area of computer science I wanted to focus on. Throughout the program, I worked with different programming languages, databases, software development tools, and computer science concepts. Some of the areas I worked with included Java, Python, C++, MongoDB, JavaScript, Angular, Node.js, and other technologies. Completing my ePortfolio helped me look back at some of that work and see how much I have learned since the beginning of the program.

One of the biggest things I learned is that computer science is not just about writing code. There is a lot of planning, testing, problem solving, communication, and security involved in creating a good solution. A program can work and still have problems if it is difficult to understand, not secure, inefficient, or does not meet the needs of the person using it. My coursework helped me understand that I need to think about the whole problem instead of only thinking about getting the code to run.

My experience with software engineering helped me understand how important organization and planning are. In CS465, I worked on the Travlr Getaways project using technologies such as Angular, Express, Node.js, and MongoDB. This project gave me experience working with a full-stack application and helped me understand how the different parts of an application connect with each other. I learned about the front end, back end, routes, data, and how information moves through an application. Working on this project also helped me become more comfortable looking at an existing project and thinking about how it could be improved instead of starting everything from the beginning.

Another important part of my program was learning about algorithms and data structures. In CS370, I worked with reinforcement learning and the Treasure Hunt project. I learned about Q-learning and how an agent can learn by making decisions and receiving rewards or penalties. This was different from some of the programming projects I had done before because I had to think about how the system was learning and how changes to the training process could affect the results. For my enhancement, I added tracking that could show information such as the number of steps used in each episode, exploration and exploitation activity, epsilon information, and average steps. This gave me a better way to look at what was happening during training instead of only looking at the final result.

Databases were another major part of my computer science education. In CS340, I worked with MongoDB and Python to create an animal shelter application. The project helped me understand

how applications communicate with a database and how data can be searched and organized. For my enhancement, I improved the AnimalShelter class by adding reusable search options for things such as animal type, breed, age range, and combined filters. This was useful because it made the database code more flexible instead of only having one basic way to search for information.

Security is also something I learned to take seriously during the program. In my secure coding coursework, I learned about problems such as SQL injection, buffer overflows, input validation, memory protection, assertions, and exception handling. I also worked with security standards and tools used to identify possible problems in code. These experiences changed the way I look at software because I now think more about what could go wrong and how a program could be misused. Security is something I want to continue improving as I move into the computer science field.

Communication was another skill I developed throughout the program. Computer science work is not always done by one person, and even when someone is working individually, they still need to communicate with users, managers, coworkers, or other stakeholders. My projects required me to explain what my code was doing, document my work, and make changes based on feedback. Creating the code review for this capstone also gave me practice explaining an existing project and discussing why I thought certain improvements were needed.

Working with other people is something I also understand better now. Some of my projects were individual school projects, but the program showed me why collaboration is important in the computer science field. Different people can look at the same problem and come up with different solutions. A good developer needs to be able to listen to those ideas and explain their own ideas clearly. I learned that being able to communicate with people who may not have the same technical background is just as important as knowing how to write code.

The projects in this ePortfolio were selected because they show different parts of my computer science education. The Travlr Getaways project represents my experience with software design and full-stack development. The Treasure Hunt project represents my work with algorithms, reinforcement learning, and tracking the results of an algorithm. The Animal Shelter project represents my database experience using Python and MongoDB. Together, these projects give a better picture of my skills than one project would by itself.

The enhancement process was also important because I was not simply submitting the projects the same way I originally completed them. I had to go back and look at my previous work and think about what could be improved. This made me notice things that I probably would not have noticed when I was first completing the projects. I also learned that improving an existing project can sometimes be harder than starting something new because I have to understand how the original code works before making changes.

Completing the Computer Science program has helped me become more confident with programming, databases, software development, algorithms, and security. It has also helped me become better at troubleshooting and breaking larger problems into smaller steps. The projects in this portfolio show some of that growth and give me something I can use when applying for jobs in the future.

My goal is to continue building my technical skills while gaining real-world experience. I want employers to see this portfolio as evidence that I can learn new technologies, work through problems, improve existing software, and explain what I am doing. I know there is still a lot for me to learn, but completing this degree and creating this portfolio has given me a strong foundation to continue growing in the computer science field.